

# Statistical Analysis Plan: Palmer Penguins

## PHB 243b Project 1

[Your Names Here]

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# 1 Introduction

## 1.1 Background and Rationale

[Your background text here]

## 1.2 Research Questions

**Primary Research Question:**

[Your primary question here]

**Secondary Questions (if any):**

1. [Secondary question 1]
2. [Secondary question 2]

## 1.3 Hypotheses

[Your hypotheses here]

# 2 Data Description

## 2.1 Data Source

[Your data source description here]

**Citation:** Horst AM, Hill AP, Gorman KB (2020). palmerpenguins: Palmer Archipelago (Antarctica) penguin data. R package version 0.1.0.

## 2.2 Study Population

[Your study population description here]

## 2.3 Sample Size

Table 1: Sample sizes by species

| species   | n_total | n_complete |
|-----------|---------|------------|
| Adelie    | 152     | 146        |
| Chinstrap | 68      | 68         |
| Gentoo    | 124     | 119        |

[Your interpretation of sample sizes here]

## 3 Variables

### 3.1 Primary Outcome

| Attribute       | Description                |
|-----------------|----------------------------|
| Variable name   | [e.g., body_mass_g]        |
| Definition      | [e.g., Body mass in grams] |
| Type            | [Continuous/Categorical]   |
| Units           | [e.g., grams]              |
| Expected range  | [e.g., 2700-6300 g]        |
| Transformations | [e.g., None / Log]         |

### 3.2 Primary Predictors

#### 3.2.1 Predictor 1: [Variable Name]

| Attribute     | Description                  |
|---------------|------------------------------|
| Variable name | [e.g., flipper_length_mm]    |
| Definition    | [e.g., Flipper length in mm] |
| Type          | [Continuous/Categorical]     |
| Rationale     | [Why include?]               |

#### 3.2.2 Predictor 2: [Variable Name]

| Attribute     | Description |
|---------------|-------------|
| Variable name |             |
| Definition    |             |
| Type          |             |

| Attribute | Description |
|-----------|-------------|
| Rationale |             |

### 3.2.3 Predictor 3: [Variable Name]

| Attribute     | Description |
|---------------|-------------|
| Variable name |             |
| Definition    |             |
| Type          |             |
| Rationale     |             |

## 3.3 Effect Modifiers / Covariates

[Your effect modifiers/covariates here]

## 4 Statistical Methods

### 4.1 Descriptive Statistics

**Continuous variables:** [mean, SD, median, IQR, range]

**Categorical variables:** [frequencies, percentages]

**Stratification:**

- ☐ Overall
- ☐ By species
- ☐ By sex
- ☐ Other: \_\_\_\_\_

### 4.2 Missing Data

Table 6: Variables with missing values

| Variable          | Missing |
|-------------------|---------|
| bill_length_mm    | 2       |
| bill_depth_mm     | 2       |
| flipper_length_mm | 2       |
| body_mass_g       | 2       |
| sex               | 11      |

**Strategy:** [Your strategy and justification here]

## 4.3 Primary Analysis

### 4.3.1 Model Specification

**Model 1:** Simple linear regression

$$\text{body\_mass} = \beta_0 + \beta_1 \times \text{flipper\_length} + \epsilon$$

**Model 2:** Multiple regression

$$\text{body\_mass} = \beta_0 + \beta_1 \times \text{flipper} + \beta_2 \times \text{bill\_length} + \epsilon$$

**Model 3:** Species-adjusted

[Your model equation here]

**Model 4:** With interactions

[Your model equation here]

### 4.3.2 Model Comparison Criteria

- ☐ R-squared / Adjusted R-squared
- ☐ RMSE
- ☐ AIC / BIC
- ☐ F-tests for nested models

## 4.4 Significance Level

- **Alpha:** [e.g., 0.05]
- **Multiple comparison adjustment:** [None / Bonferroni / FDR]

## 4.5 Diagnostic Checks

| Assumption         | Diagnostic Method   | If Violated |
|--------------------|---------------------|-------------|
| Linearity          | Residual vs. fitted | Transform   |
| Normality          | Q-Q plot            |             |
| Homoscedasticity   | Scale-location plot |             |
| Independence       | Study design        |             |
| Influential points | Cook's distance     |             |
| Multicollinearity  | VIF                 |             |

## 5 Planned Results

### 5.1 Table 1: Sample Characteristics

| Characteristic             | Adelie | Chinstrap | Gentoo | Overall |
|----------------------------|--------|-----------|--------|---------|
| <b>n</b>                   |        |           |        |         |
| <b>Bill length (mm)</b>    |        |           |        |         |
| Mean (SD)                  |        |           |        |         |
| <b>Bill depth (mm)</b>     |        |           |        |         |
| Mean (SD)                  |        |           |        |         |
| <b>Flipper length (mm)</b> |        |           |        |         |
| Mean (SD)                  |        |           |        |         |
| <b>Body mass (g)</b>       |        |           |        |         |
| Mean (SD)                  |        |           |        |         |
| <b>Sex, n (%)</b>          |        |           |        |         |
| Female                     |        |           |        |         |
| Male                       |        |           |        |         |

### 5.2 Table 2: Correlation Matrix

|             | Bill Length | Bill Depth | Flipper | Body Mass |
|-------------|-------------|------------|---------|-----------|
| Bill Length | 1.00        |            |         |           |
| Bill Depth  |             | 1.00       |         |           |
| Flipper     |             |            | 1.00    |           |
| Body Mass   |             |            |         | 1.00      |

### 5.3 Table 3: Model Comparison

| Term             | Model 1 | Model 2 | Model 3 | Model 4 |
|------------------|---------|---------|---------|---------|
| Intercept        |         |         |         |         |
| Flipper length   |         |         |         |         |
| Bill length      |         |         |         |         |
| Bill depth       |         |         |         |         |
| Species          |         |         |         |         |
| <b>R-squared</b> |         |         |         |         |
| <b>RMSE</b>      |         |         |         |         |
| <b>AIC</b>       |         |         |         |         |

## 5.4 Planned Figures

### 5.4.1 Figure 1: Species Distribution

- **Type:** Bar chart
- **X-axis:** Species
- **Y-axis:** Count
- **Purpose:** Show sample composition

### 5.4.2 Figure 2: Predictor vs. Outcome

- **Type:** Scatter plot
- **X-axis:** Flipper length
- **Y-axis:** Body mass
- **Color:** Species
- **Purpose:** Visualize primary relationship

### 5.4.3 Figure 3: Correlation Heatmap

- **Type:** Heatmap
- **Variables:** All numeric
- **Purpose:** Show variable relationships

### 5.4.4 Figure 4: Model Diagnostics

- **Type:** 2x2 panel (residuals, Q-Q, scale-location, leverage)
- **Purpose:** Assess model assumptions

## 6 Sensitivity Analyses

| Analysis              | Rationale                | Expected Impact |
|-----------------------|--------------------------|-----------------|
| Exclude outliers      | Test robustness          |                 |
| Multiple imputation   | Compare to complete case |                 |
| Stratify by species   | Check pooling validity   |                 |
| Log-transform outcome | Address non-normality    |                 |

## 7 Limitations

### 7.1 Scope and Generalizability

[Results specific to Palmer Archipelago population]

## 7.2 Causal Inference

[Observational data, cannot establish causation]

## 7.3 Data Limitations

[Limited time period 2007-2009, potential measurement error]

## 7.4 Methodological Limitations

[Model assumptions, variable selection]

## 8 Team Responsibilities

| Task                   | Responsible | Deadline |
|------------------------|-------------|----------|
| Data cleaning          |             |          |
| Descriptive statistics |             |          |
| Primary analysis       |             |          |
| Figures                |             |          |
| Write-up               |             |          |
| Review/editing         |             |          |

## 9 References

1. Gorman KB, Williams TD, Fraser WR (2014). Ecological sexual dimorphism and environmental variability within a community of Antarctic penguins (genus *Pygoscelis*). *PLOS ONE* 9(3): e90081.
2. Horst AM, Hill AP, Gorman KB (2020). palmerpenguins: Palmer Archipelago (Antarctica) penguin data. R package.
3. [Additional references]

## 10 Appendix: Session Information

```
sessionInfo()
```



Running under: macOS Tahoe 26.2

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LA
```

[1] en\_US.UTF-8/en\_US.UTF-8/en\_US.UTF-8/C/en\_US.UTF-8/en\_US.UTF-8

```
tzcode source: internal
```

```
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
[10] tibble_3.3.0      ggplot2_4.0.1      tidyverse_2.0.0
```

```
[33] pkgconfig_2.0.3      htmltools_0.5.9
```